

WASP-43b

R-0340

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_260209 · created 2026-07-18T06:33:59+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2461080.8341 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.184 +/- 0.027
Transit depth (Rp/Rs)^2	3.37 +/- 1.0 [%]
Orbital Inclination (inc)	79.26 +/- 1.64
Airmass coefficient 1 (a1)	1.123 +/- 0.013
Airmass coefficient 2 (a2)	-0.079 +/- 0.017
Scatter in the residuals of the lightcurve fit is	1.12 %
Best Comparison Star	None
Optimal Aperture	4.03
Optimal Annulus	10.48
Transit Duration (day)	0.04 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.184 $\pm$ 0.027 vs 0.1594 (deviation 0.91 σ)
Duration fit vs published	0.04 d vs 0.0512 d (deviation 1.02 σ)
Mid-time O–C	7.1 min
Depth SNR	6.8
Residual scatter	1.12 %

Light curve

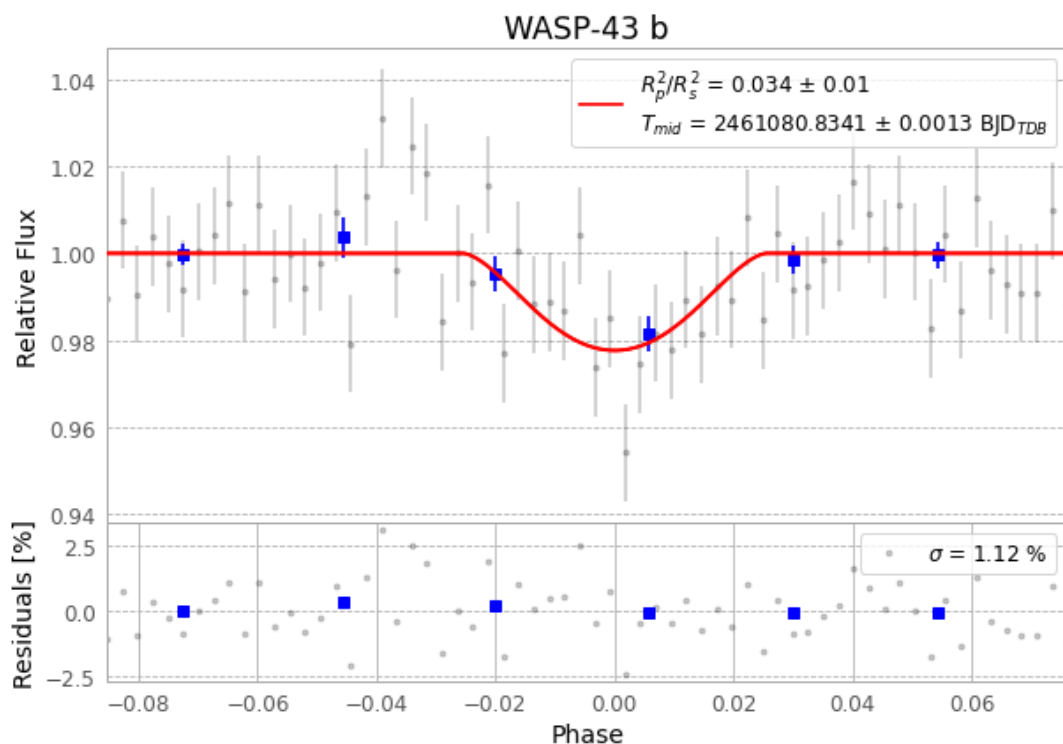


Figure 1 — FinalLightCurve_WASP-43 b_2026-02-08.png (SHA-256 00897090559dbaa2...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [452, 175], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1e7c1df15c8bda60...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8b944da

Data provenance

64 FITS frames (42 MB), WASP-43260209061215.FITS → WASP-43260209092115.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-260209.log (ceac45b10d64...), FinalLightCurve_WASP-43 b_2026-02-08.png (00897090559d...), FinalLightCurve_WASP-43 b_2026-02-08.csv (08b0937f5da0...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 06:33Z.